



CHECK-IN

KUISC - NETWORKING FUNDAMENTALS
WEDNESDAY, SEPTEMBER 16 AT 6:00PM



<https://cglink.me/2rx/c377790915102241>



NETWORKING BASICS + WIRESHARK CTF

Network Traffic Analysis: Meeting 3

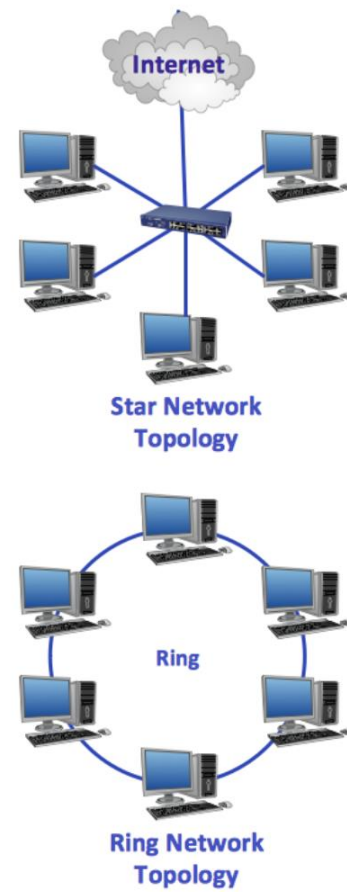
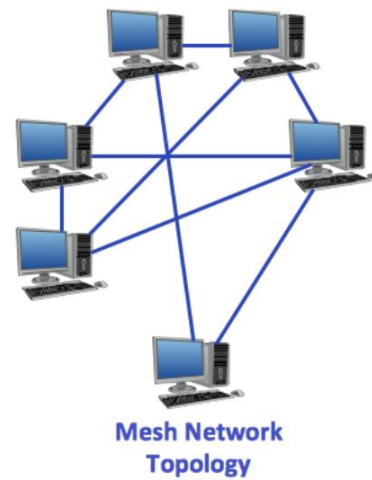
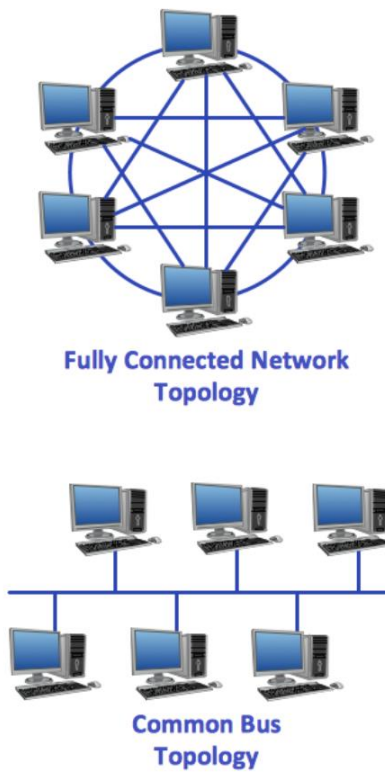
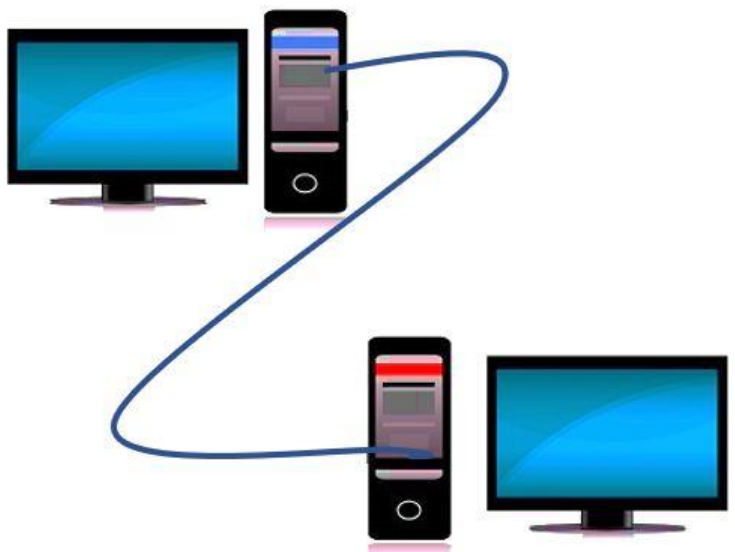
What is a network?

HOST <-> LINK <-> HOST

The simplest possible network: two nodes and a link between them

- **Nodes** - End systems (computers, phones, printers) plus the switches and routers that move data between them.
- **Links** - The physical or wireless connection between nodes: copper, fiber, radio
- **Network** - Two or more nodes connected by a sequence of links.

Question: Which network does our lab use?



Switches vs. routers

Switch

- **Lives inside one network** - Connects hosts on the same local network (a LAN).
- **Reads MAC addresses** - Forwards frames to the right port based on hardware address.
- **Doesn't leave the building** - Traffic between two switches never crosses onto another network.

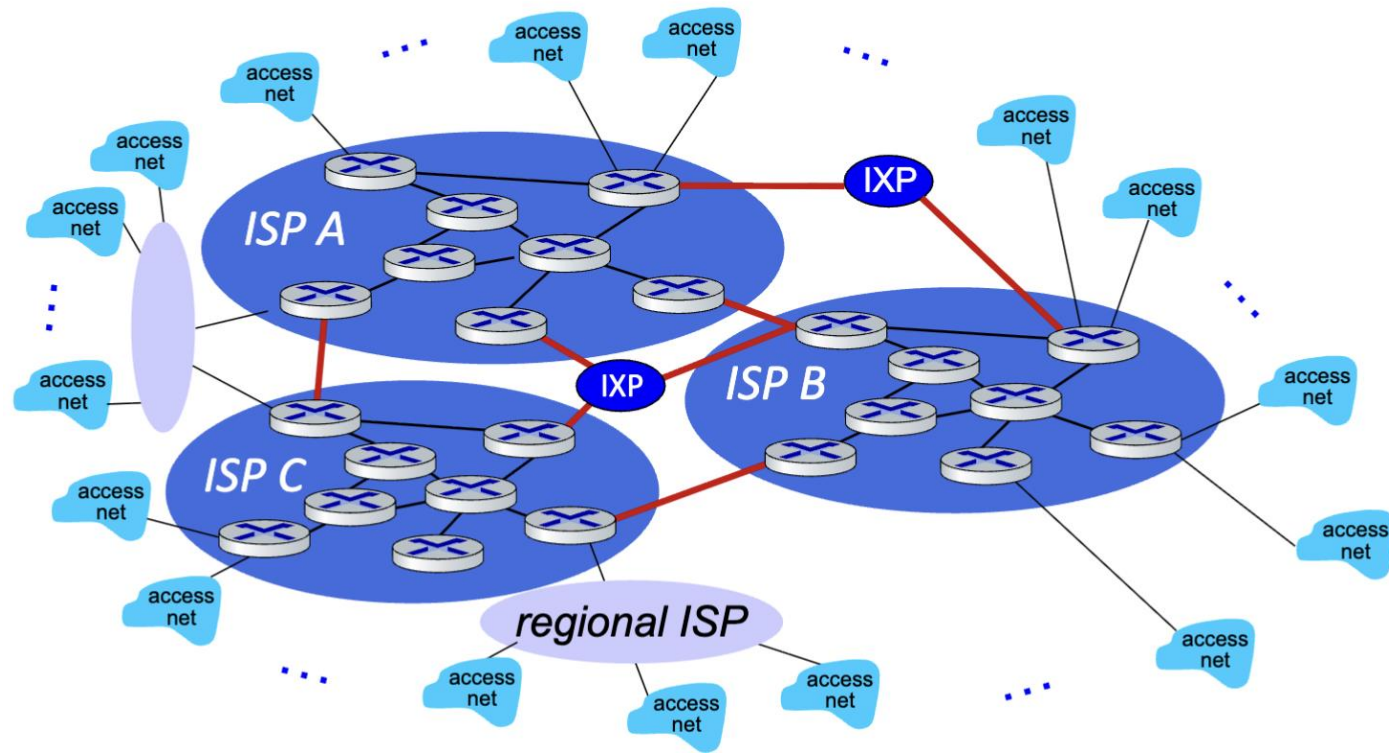


Router

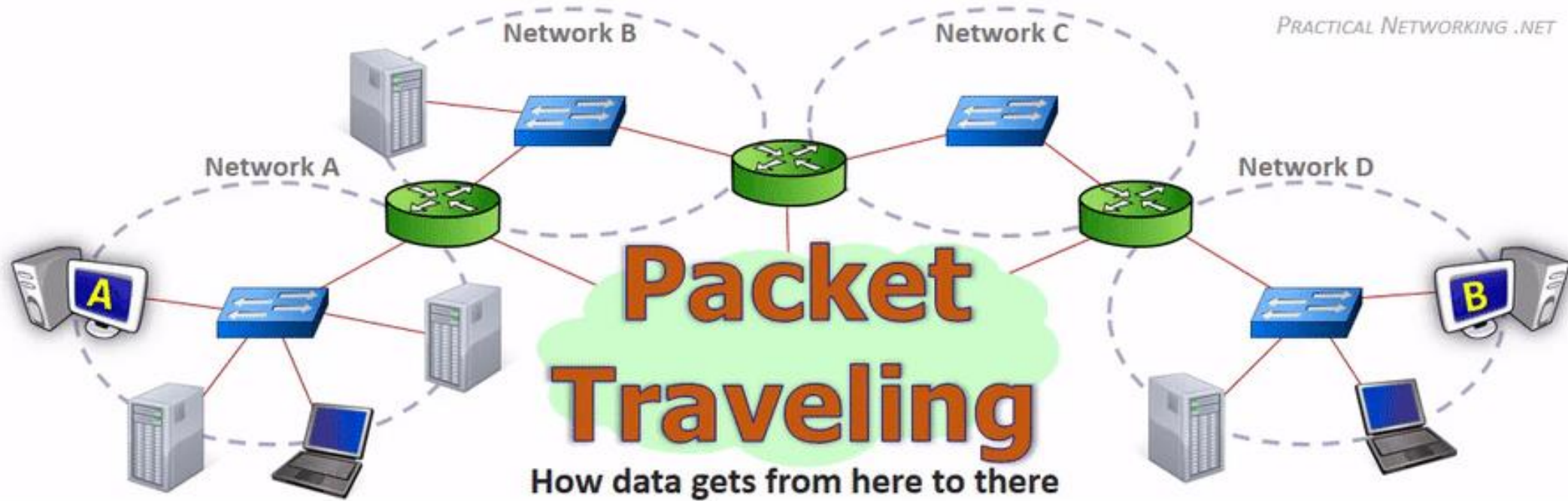
- **Connects networks together** - Sits at the boundary between one network and the next.
- **Reads IP addresses** - Decides which direction to send a packet based on its destination.
- **Makes routing decisions** - Every hop across the internet passes through a router.



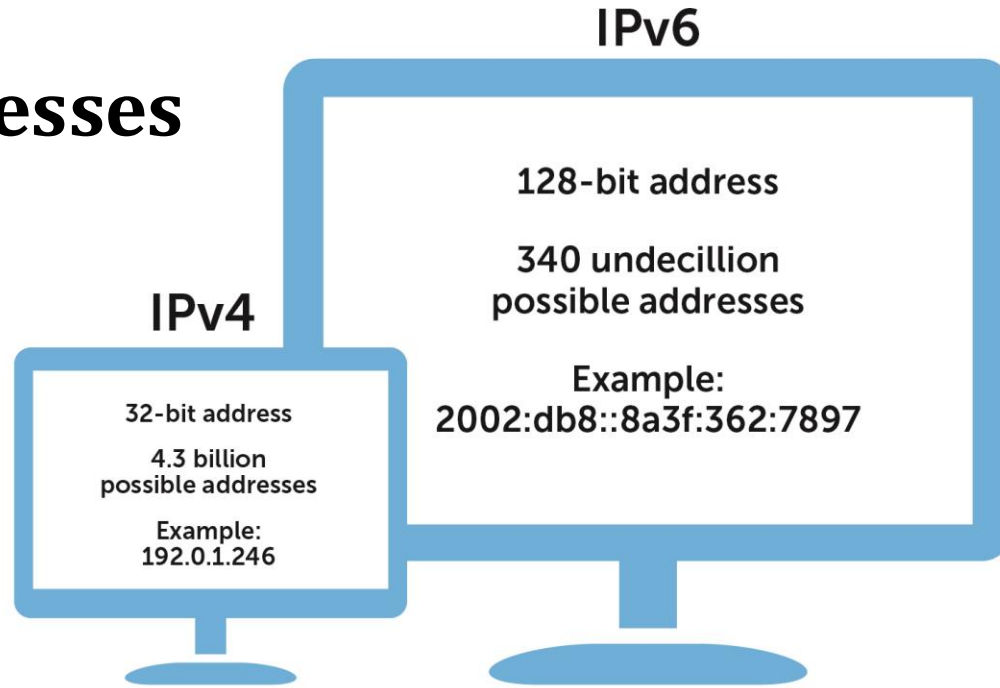
The Internet: a network of networks



How do we traverse the internet?



IP addresses



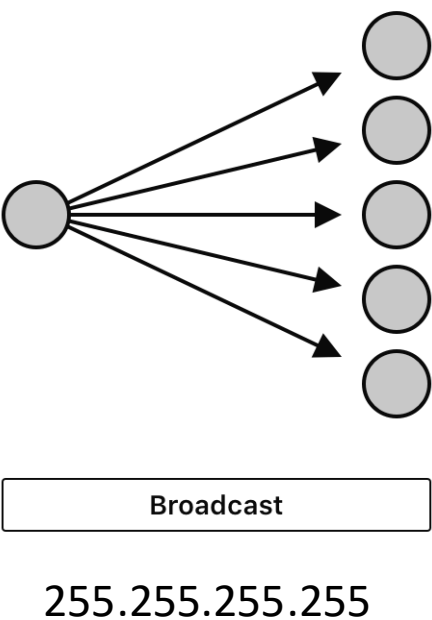
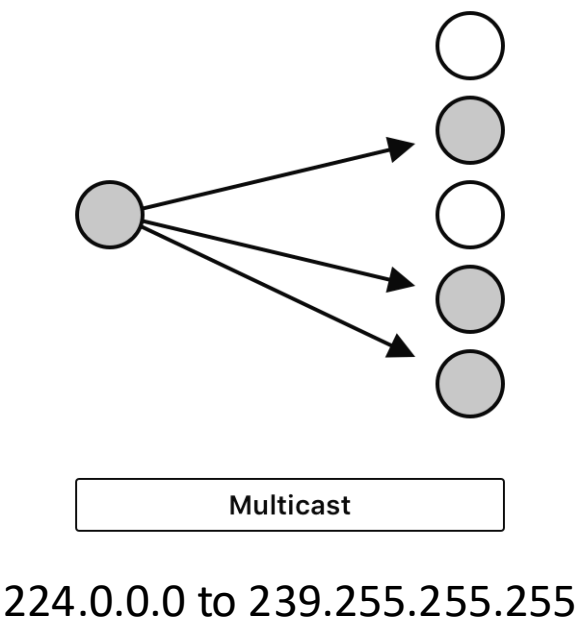
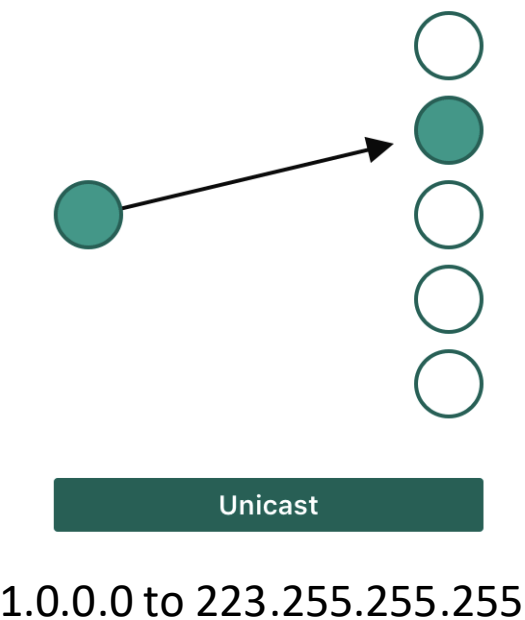
IPv4 Header				
Version	IHL	Type of Service	Total Length	
Identification			Flags	Fragment Offset
Time to Live	Protocol	Header Checksum		
Source Address				
Destination Address				
Options				Padding

Legend		Field's Name Kept from IPv4 to IPv6
		Fields Not Kept in IPv6
		Name and Position Changed in IPv6
		New Field in IPv6

IPv6 Header			
Version	Traffic Class	Flow Label	
Payload Length		Next Header	Hop Limit
Source Address			
Destination Address			

- **Two versions: IPv4 and IPv6** - IPv4 is most commonly used, but does not have as many possible addresses as IPv6. Packet composition differs between the two versions. For today, we will only interact with IPv4, sorry Alex :)
- **IP identifies a device on a network** - Routers read the destination address to decide where to send the packet next.
- **Private vs. Public** - 10.x, 172.16-31.x, and 192.168.x only work inside a local network; a public IP is how the internet finds you.

Who gets the packet?



© TechTerms.com

MAC addresses: the physical hop

AA : BB : CC : 11 : 22 : 33

Six pairs of hex digits, burned into the network card

- **Assigned by the manufacturer** - The first half of the address identifies who made the card. MAC address **doesn't change**.
- **Each Frame gets a source and destination MAC** – Source is the where the packet is being sent from, destination is the MAC of the next node in the path.
- **Only good for one hop** - It gets a frame from your laptop to your router. Subsequent hops at the router strip the MAC header and replace it with a new source and destination MAC.

Ports: which app gets the data



source IP : source port → destination IP : destination port

- **A port is a number, 0-65535** - It tells the receiving computer which app or service the data is for.
- **Servers listen on ports** - 80 for HTTP, 443 for HTTPS, 53 for DNS, or others corresponding to apps and services. The client picks a random port for itself to send packet.
- **IP plus port defines a single conversation** - This allows for multiple conversations to occur at once.

TCP vs. UDP: two ways to send data

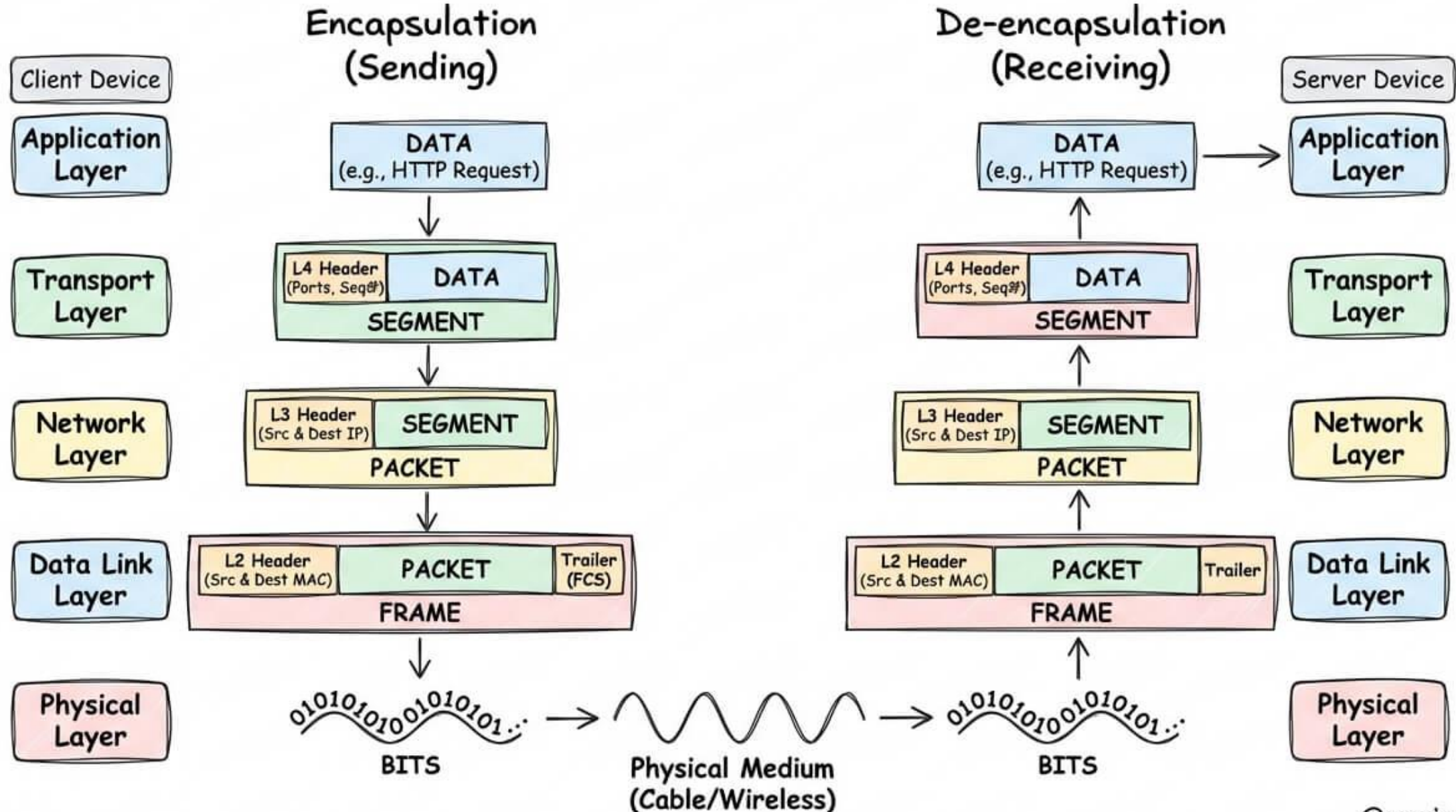
TCP

- **Connection first** - A three-way handshake (SYN, SYN-ACK, ACK) happens before any data moves.
- **Reliable** - Lost packets get detected and retransmitted automatically.
- **Ordered** - Data arrives in the same order it was sent, every time.
- **Used for** - HTTP, HTTPS, FTP, SSH, email, and anything that needs to arrive intact.

UDP

- **No handshake** - No setup or teardown, no waiting around.
- **No guarantees** - A lost packet stays lost; nothing resends it automatically.
- **No ordering** - Packets can arrive out of sequence, or not arrive at all.
- **Used for** - DNS, video calls, live streaming, gaming. Places where speed matters over certainty.

Data Encapsulation & Packet Flow



Why look at packets?



Attackers can't hide from the wire

Logins, C2 beacons, and stolen data all have to cross the network. Every bit of it can be captured.



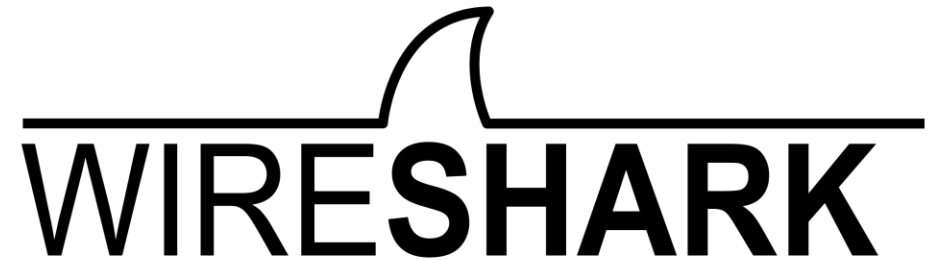
It's how the internet works

HTTP, DNS, TCP: Wireshark shows you the real protocols running underneath every site and app.



Core incident response skill

Responders reconstruct what happened by reading traffic.

The Wireshark logo, featuring a stylized shark fin above the word "WIRESHARK" in a bold, sans-serif font.



CHALLENGE TIME

How to access Wireshark

Bring your own laptop, or use a lab computer

Lab Login Details – Username: kali, Password: toor

Install Wireshark:

`wireshark.org/download.html`

Pcap files handed out at the meeting:

`basic-http.pcap` `dns-basics.pcap` `incident.pcap`

Task: Complete the Wireshark CTF

- 1 Grab a Cheat Sheet + Question Bank
- 2 Open each pcap in Wireshark and work through the questions
- 3 First to submit all correct answers wins a prize



TRIPLE CHOCOLATE

540 cal





Questions?
